Agricultural Water Management 199 (2018) 48–60

Contents lists available at ScienceDirect

Agricultural Water Management

journal homepage: www.elsevier.com/locate/agwat

Review

Determining water use efficiency of wheat and cotton: A meta-regression analysis

Yubing Fan[a,*], Chenggang Wang[b], Zhibiao Nan[c]

[a] Texas A&M AgriLife Research, Vernon, TX 76384, USA
[b] Department of Agricultural and Applied Economics, Texas Tech University, Lubbock, TX 79409, USA
[c] State Key Laboratory of Grassland Agro-ecosystems, College of Pastoral Agriculture Science and Technology, Lanzhou University, Lanzhou, Gansu 730020, China

ARTICLE INFO

Article history:
Received 10 June 2016
Received in revised form
21 November 2017
Accepted 4 December 2017
Available online 16 December 2017

Keywords:
Water use efficiency
Micro irrigation
Farm best management practices
Wheat
Cotton
Meta-regression analysis
Publication bias

ABSTRACT

A great challenge for agricultural production is to produce more food with less water, which can be possibly achieved by increasing crop water use efficiency (WUE). This study systematically reviewed 52 cases from 49 empirical studies with field experimental results on wheat and cotton. This research investigated yield-water use relations under both furrow and micro irrigation systems, compared optimal water use to achieve maximum WUE and maximum yield, calculated water saving potentials under various scenarios, and evaluated the effects of influential factors using meta-regression analysis. The results suggest that to achieve maximum WUE rather than maximum yield, water use for wheat can be reduced by 30.4% with a grain yield decrease of 14.8%, and water use for cotton can decrease by 51.4% with a yield reduction of 51.7%. Compared with furrow irrigation, micro irrigation reduces wheat water use by 22.7% and increases yield by 36.7%. While for cotton, micro irrigation reduces water use by 36.8% and decrease yield by 21.4%. Under the scenario of a 10% yield reduction, water use decreases by 25% for wheat and by 20–22% for cotton. Compared with maximum yield, other yield levels reduce water use by 2–15% on average for wheat, and by 15–17% for cotton. Achieving maximum WUE reduces water use by 14–31% compared with other sub-optimal WUE levels. The meta-regression analysis showed adoption of micro irrigation systems, and farm management practices on soil and water significantly improved wheat and cotton WUE. Assessments of the publication selection bias and genuine effects illustrate the application of weighted least squares in conducting meta-regression analysis.

© 2017 Elsevier B.V. All rights reserved.

Contents

1. Introduction 49
2. Literature review 49
   2.1. Crop water use efficiency and farm management practices 49
   2.2. Meta-regression analysis 50
3. Data and methods 50
   3.1. Meta-data collection 50
   3.2. Defining water use efficiency 51
   3.3. Independent variables 52
      3.3.1. Measures of estimate's precision 52
      3.3.2. Characteristics of the data 52
      3.3.3. Adoption of micro irrigation systems 52
      3.3.4. Climate of the study area 52
      3.3.5. Farm management practices 52

* Corresponding author.
E-mail addresses: fanyubing@126.com, yubing.fan@ag.tamu.edu (Y. Fan), chenggang.wang@ttu.edu (C. Wang), zhibiao@lzu.edu.cn (Z. Nan).

https://doi.org/10.1016/j.agwat.2017.12.006
0378-3774/© 2017 Elsevier B.V. All rights reserved.

3.3.6. Characteristics of the empirical study[5] 52
3.4. Econometric issues 53
3.5. Econometric models for meta-regression analysis 53
4. Results 54
4.1. Meta-data overview 54
4.2. Yield-ET relations of wheat and cotton 54
4.3. Water saving potentials 55
4.4. WUE of wheat and cotton 56
4.5. Results of meta-regression analysis 56
5. Concluding remarks 58
Acknowledgements 59
Appendix A. Supplementary data 59
References 59

1. Introduction

With the rapid growth of the world population, fresh water becomes increasingly scarce across the globe. Agriculture is the largest water-consuming sector and water shortage is a principal limiting factor for crop production. The problem is most acute in regions where rainfall is limited or highly variable. For example, the annul precipitation is less than 200 mm in arid areas of Hexi Corridor, Gansu Province, northwestern China (Huang et al., 2012; Fan et al., 2014); it's only about 105 mm in Xinjiang Province, China, and a majority occurs from June to August (Kang et al., 2012); in North China Plains, although the mean precipitation is 500–600 mm, the annual crop evapotranspiration (ET) of 800–900 mm considerably exceeds the precipitation (Jin et al., 1999; Liu et al., 2002). Therefore, to offset water deficit and maintain a high crop grain yield in those areas, agricultural production relies heavily on irrigation. In addition, water scarcity is due to lack of surface water in some areas of the world, such as northwestern China and North China Plains, Southern Texas High Plains of U.S., and parts of Uzbekistan, Syria, Turkey and India (Fan et al., 2014; Murray et al., 2012). In such regions, groundwater becomes a primary source for agricultural irrigation, resulting in persistent declining of groundwater levels and considerably large zones of groundwater depression (Bordovsky et al., 1999; Du et al., 2006; Ibragimov et al., 2007; Liu et al., 2011; Oweis et al., 2011; Singh et al., 2010; Yazar et al.,2002).

Limited availability of irrigation water requires fundamental changes in irrigation management and promotes application of water saving techniques. Traditionally, furrow, flood and basin irrigations are the common methods. These irrigation systems tend to over-irrigate croplands, resulting in a waste of water and low water use efficiency (WUE) (Yazar et al., 2002b). Micro irrigation systems (e.g., drip emitters, drip tape, spray, and sprinklers), either spraying water to plants or dripping near their root zone, save 30–70% of the irrigation water and has gained increasing popularity in irrigated agricultural production (Ibragimov et al., 2007; Kang et al., 2012; Yazar et al., 2002b). With unique agronomic and economic advantages, micro irrigation also shows a potential of precisely applying water and chemicals across croplands which reduces labor and energy inputs (Gärdenäs et al., 2005; Levidow et al., 2014). Much research regarding the irrigation effects on cotton demonstrated that micro irrigation systems led to improved yields and more efficient water use than traditional methods (Bucks et al., 1988; Hodgson et al., 1990; Mateos et al., 1991). Comparative studies between micro and traditional irrigation systems have revealed a significant increase of grain yields, harvest index and crop water use efficiency (Cetin and Bilgel, 2002; Ibragimov et al., 2007; Schneider and Howell, 2001; Yazar et al., 2002a), provided that the irrigation systems are properly designed, managed, operated and maintained.

Meta-regression analysis (MRA) is a statistical model used to analyze data points obtained from separate empirical studies (Phillips, 1994; Stanley and Jarrell, 1989). MRA has the advantage of being able to systematically account for a complex set of potential factors that may influence some dependent variable under concern, and to draw conclusions from the analysis of literature (Loomis and White, 1996; Smith and Kaoru, 1990; Stanley, 2001). To the best of authors' knowledge, however, a comprehensive meta-regression analysis of the relations between WUE and irrigation systems as well as other farm management practices has not yet been conducted, and the purpose here is to fill this gap.

Based on a pooled database obtained from empirical studies, this paper compares WUE with estimation of production functions under both furrow and micro irrigation systems, and evaluates WUE of wheat and cotton using meta-analytical techniques. Specifically, the objectives are: (1) to investigate minimum and maximum values of WUE for wheat and cotton, explore potential water saving under various scenarios, and provide a comparable benchmark for future studies, (2) to explore the potential relationship between WUE and various influential factors, especially effects of micro irrigation and farm management practices on crop, soil, water and fertilizer[1] in improving WUE of wheat and cotton, and (3) to evaluate the application of MRA in synthesizing studies of agricultural water management, and examine publication selection bias and applicability of multiple econometric models.

2. Literature review

2.1. Crop water use efficiency and farm management practices

Higher WUE and/or higher yield can be achieved by applying various farm best management practices (BMPs) (Levidow et al., 2014), for instance, no/minimum/rotational tillage, straw/film mulching, etc. The research by Hou et al. (2012) showed that compared to the conventional tillage, the rotational tillage significantly improved soil moisture status, increased the amount of soil

[5] The coding of the selected studies originally included several other variables, including precipitation, more categories of climates, and fertilizer application rate. Given the primary goal of this study and multicollinearity concerns, these variables were finally excluded from the analysis to ease the analysis and interpretation of the results.

[1] Crop management practices include crop rotations, annual double cropping, planting catch crops, better crop varieties, etc. (Ingram et al., 2012). Soil management practices refer to alternative/rotational tillage, no/minimum tillage, terracing maintenance, sowing (Ingram et al., 2012; Jalota et al., 2008). Water management practices include investigation of crop water requirements and crop evapotranspiration during various growing stages, scientific irrigation scheduling, amount of effective rainfall, and supplementary irrigation (Cayci et al., 2009; Kang et al., 2012). Fertilizer management practices include investigating optimal combination of N, K, and P fertilizer, and varying frequencies and amount of fertilizer application during multiple growing stages (Komilov et al., 2003; Qin et al., 2016).

water stored during the growing season of wheat, and resulted in a 9.6–10.7% increase of wheat yield and a 7.2–7.7% increase of WUE. Cayci et al. (2009) examined the effect of rotation of winter wheat with five crops and fallow, and found the highest crop water use was determined in the preceding fallow treatment and the WUE values of winter wheat were higher in the preceding spring lentil treatment than in the preceding fallow treatment. Many other empirical studies investigated crop WUE for varying water amounts applied and/or under different irrigation systems, and analyzed effects of climatic conditions, fertilizer levels, and mulching patterns (Albrizio et al., 2010; Chakraborty et al., 2008; Tolk et al., 1999; Xie et al., 2005; Zhao et al., 2012). The adoption of mulching and fertilizer management has been reported to achieve higher WUE (Chakraborty et al., 2008; Xie et al., 2005). Field mulching curtails evaporation from the soil and holds water near the root zone (Fan et al., 2014), and appropriate fertilizer application contributes to crop growth and better performance of photosynthesis (Albrizio et al., 2010). In addition, residue management practices keeping straws in field improves the carbon sequestration in the long term, which benefits plants absorbing moisture from soil (Luo et al., 2010).

Furthermore, as water use efficiency can be effectively improved by reducing soil evaporation and/or increasing grain yield (Baumhardt and Jones, 2002; Howell et al., 2004), some other farm management strategies can be utilized in crop production. For instance, it is an effective way to reduce water use by leaving wheat under a slight drought stress in early growing stages and much research has been reported to show that a certain degree of soil water deficit can be kept during the stages at which crops are not sensitive to drought stresses (Li et al., 2005; Schneider and Howell, 2001; Singh et al., 2010; Tang et al., 2010). Too much irrigation water can result in low crop WUE, and effective irrigation with less water leads to a higher yield and WUE (Zhang and Oweis, 1999; Zhang et al., 2004). Thus, it is important and necessary to explore and interpret the relations between WUE, yield, and ET, as well as the effects of crop, soil, water, and fertilizer management practices, and other influential factors.

2.2. Meta-regression analysis

With increasing application of MRA in the fields of health science, education and psychology, meta-analysis techniques have been introduced into economic studies (Stanley, 2001; Stanley and Jarrell, 1989) and explored by economists as a possible basis for assessing the non-use value and nonmarket benefit transfer (Bateman and Jones, 2003; Bergstrom and Taylor, 2006; Johnston et al., 2003; Rosenberger and Stanley, 2006; Van Houtven et al., 2007), as well as for modeling the relationships of non-use values and willingness to pay (WTP) for environmental goods, and estimating the variability in WTP (Florax et al., 2005; Johnston et al., 2003; List and Gallet, 2001; Santos, 1998). For instance, Boyle et al. (1994) evaluated the willingness to pay for ground water contamination using meta-analysis. Loomis and White (1996) meta-analyzed annual WTP for 18 different rare and endangered species, and demonstrated meaningful estimates of anthropocentric benefits of preserving these species by means of contingent valuation. MRA has also been applied to analyze farm productivity as affected by socio-economic factors (Phillips, 1994; Stanley et al., 2008).

In the fields of environmental and natural resource economics, agronomy and farm management, some studies conducting meta-analysis have been published. A summary of some recent meta-analysis studies is presented in Table A1 of Appendix A. There is some similar research on agricultural production and crop water use. For instance, Bravo-Ureta et al. (2007) meta-analyzed the farm level technical efficiency using 167 studies conducted around the world and found that varying estimates of the technical efficiency obtained were due to the differences of empirical models employed and variations at the country and continent levels. Both Pittelkow et al. (2015) and Zhao et al. (2016) analyzed the effects of no-till farming. Focusing on farm productivity, Pittelkow et al. (2015) identified 678 studies representing 50 crops and 63 countries, and found that the best performance of no-till was for rainfed agriculture in dry climates given the evidence that the yields were equal to or higher than that of conventional tillage farming. With a comprehensive analysis of 39 studies in China, Zhao et al. (2016) pointed out that the adoption of non-tillage mitigated higher N2O emission in alkaline soils and under other conditions of fertilization and duration. In addition, three fundamental meta-analysis studies reported crop water use. Sadras (2009) compared the effects of partial root-zone drying and conventional deficit irrigation on yields of multiple crops, and confirmed that the yield per unit water was higher by applying the partial root-zone drying, and this method was also more feasible and economical. Qin et al. (2016) conducted a meta-analysis to evaluate water and nitrogen use efficiencies in citrus production and found reductions of water use and fertilizer use might increase citrus yield, WUE and nitrogen use efficiency by 10–40%. In a similar study, Qin et al. (2015) concluded that compared with no-mulching, soil mulching significantly increased yields, WUE and nitrogen use efficiency by up to 60% for wheat and corn. As shown in Table A1 of Appendix A, the meta-analysis studies mentioned above (along with some others in the table) employed different meta-analytical techniques and/or econometric models.

Notwithstanding its growing application, meta-analysis has never been free from criticism. A focal point of the critiques is that under-/over-estimation of the true effect may come from publication selection biases[2] due to a higher probability of getting published for studies with significant results, unequal quality of study design and publications, aggregation of separate studies, unequal weights for the empirical studies included, selection and comparability of moderator variables, etc. (Phillips, 1994; Stanley, 2001; Stanley, 2005; Stanley et al., 2008). Publication bias reduces the validity and reliability of meta-regression analyses, that is, distorting the estimates of measures and making empirical effects seem larger than they actually are (Rosenberger and Stanley, 2006; Stanley et al., 2008).

To some extent, the quality of a meta-analysis is basically determined by how well the authors deal with and account for the publication bias. Some associated techniques to assess/correct the publication bias include funnel plot, funnel asymmetry test (FAT), and precision-effect test (PET). In addition, the weighted least squares (WLS) models with standard error or number of observations in each study as weights are appropriate for meta-regression analysis (Stanley, 2005; Stanley et al., 2008; Stanley and Jarrell, 1989).

3. Data and methods

3.1. Meta-data collection

To compare and evaluate WUE of wheat and cotton using different irrigation methods and farm management practices, it's critical to conduct systematic and exhaustive literature searches and screening.[3] In 2012, extensive literature searches were con-

[2] Given the great interest from social scientists and promising application of MRA, a special issue in the Journal of Economic Surveys was designed in 2005, which demonstrated the research scope that MRA can address and showcased the tests that should be conducted along with MRA to cope with potential biases (see the articles by Roberts (2005), Stanley (2005), Stanley and Jarrell (1989), etc.).

[3] As the database has not been updated since 2013, instead of comprehensively reviewing WUE literature, this study focuses on investigating some general patterns

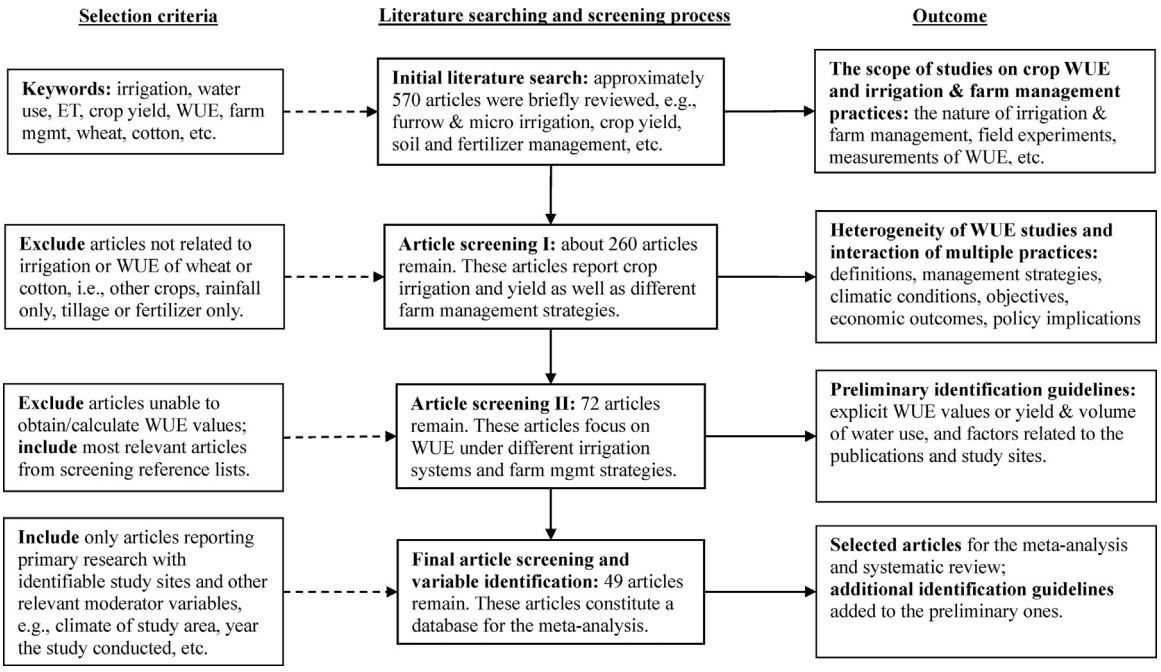


Fig. 1. Literature searching and screening process for selecting sample publications used in the meta-analysis (Adapted from Pagdee et al., 2006).

ducted within many academic literature databases and using several search engines, including Elsevier (ScienceDirect), Emerald, SpringerLink, Wiley, Google Search, Baidu Search and so forth. Some additional studies were located by scanning the reference lists of identified publications. The authors tried to identify as many empirical studies as possible, including publications, reports, thesis, and other unpublished materials.

A general procedure to choosing empirical studies for systematic reviews and meta-analysis was followed. Fig. 1 shows the literature searching and screening process. Approximately 570 articles relating to any combinations of relevant keywords, e.g., water use, crop yield, water use efficiency, farm/crop/soil/water/fertilizer management, wheat, cotton, etc. were reviewed initially. During the initial search, the scope of studies on crop WUE, and irrigation and farm management strategies were learned, including nature of irrigation and farm management practices, field experiments, measurements of WUE, etc. Subsequently, two rounds of article screening were carried out. The first screening excluded articles unrelated to farm irrigation of wheat and corn, i.e., studies focused on other crops, non-irrigation, only tillage or fertilizer applications while no information on water use, etc. Through screening and reading diverse empirical studies, the heterogeneity of WUE studies and interactions of multiple practices were understood, including definitions, multiple farm BMPs, climatic conditions and their effects, etc. The second screening excluded articles from which the authors were unable to obtain or calculate WUE values, and included some most relevant articles from scanning the reference lists. After the two rounds of article screening, 72 empirical studies focusing on WUE under different irrigation systems and using farm management strategies remained. They met some preliminary identification guidelines including articles should present crop WUE values or values of crop yield and volume of water use, as well as factors and information related to the publication and study sites. Afterwards, during the final screening, the identification guidelines were supplemented with additional information, including clear identification of climate types of the study area, the year when the study was conducted, number of observations, etc. In the end, 49 articles remained for the meta-analysis, specifically, 27 empirical studies on wheat and 25 on cotton.[4]

Although the literature searches were conducted in English and Chinese, the number of identified publications that provided complete and comparative information of wheat and cotton WUE for meta-analysis was limited. Unfortunately, many studies just focused on either determination of crop water use and yield or application of irrigation water without reporting their effects on WUE. All selected articles presented explicit values of WUE or values of crop yield and water use which facilitated the calculation of WUE. In addition, these articles provided direct or indirect information regarding all predictor and moderator (characteristics of publications) variables mentioned below. The identification process was repeated three times to ensure consistent data classification.

3.2. Defining water use efficiency

Water use efficiency can be calculated from dry grain yield per unit cropland (Y, $kg\,ha^{-1}$) divided by the amount of water consumed by the crop (ET, *mm*) to produce that yield (Ibragimov et al., 2007).

$$\mathrm{WUE} = \frac{Y}{ET} \tag{1}$$

where WUE refers to crop water use efficiency and the unit is $kg\,m^{-3}$, which can be unified with the unit $kg\,ha^{-1}\,mm^{-1}$ following:

$$WUE_{a,(kg\,m^{-3})} = WUE_{b,(kg\,ha^{-1}mm^{-1})}/10 \tag{2}$$

ET is crop evapotranspiration, and usually expressed as a depth of water (*mm*). ET consists of water from precipitation (*P*), irrigation (*I*) and soil water content change (ΔW) while excluding surface water runoff (*R*) (Tong et al., 2007):

$$ET = P + I + \Delta\mathrm{W} - R \tag{3}$$

of WUE studies in the literature in addition to estimation of water use, yield and WUE relations.

[4] Three articles reported both crops. See the notes under Table 2.

Table 1
Variables used in the meta-regression analysis and summary statistics.

Variable	Description	Unit and measurement (ranges for wheat and cotton)	Mean (S.D.[a])	
			Wheat	Cotton
WUE	Mean of crop water use efficiency (WUE) in a publication	kg m^{-3} (0.54–1.73; 0.15–1.32)	1.13 (0.29)	0.60 (0.27)
SE	Standard error of WUE in a publication	SE of WUE (0.016–0.171; 0.002–0.145)	0.058 (0.035)	0.036 (0.040)
SQRT(n)	Square root of sample size in a publication	SQRT (n) (1.732–5.657; 1.41–4.47)	3.65 (1.18)	2.880 (0.970)
YEAR	Year in which a study was started, converted to an index by subtracting 1980	Year index (2–28; 2–29)	16.81 (6.31)	20.48 (6.31)
NUM_YEAR	Number of years a study lasted	Years (1–13; 1–18)	3.59 (2.36)	3.12 (3.23)
NUM_DATA	Number of data points used to calculate the mean WUE, yield and ET	Data points (3–32; 2–20)	14.67 (8.83)	9.20 (5.97)
IRRI_MICRO	Dummy variable indicating the micro irrigation was applied	Binary (0 or 1)	0.48 (0.51)	0.48 (0.51)
CLIM_ARID	Dummy variable indicating the study area had an arid or semiarid climate	Binary (0 or 1)	0.44 (0.51)	0.76 (0.44)
MGMT_CROP	Dummy variable indicating a study focused on crop management practices	Binary (0 or 1)	0.30 (0.47)	0.32 (0.48)
MGMT_SOIL	Dummy variable indicating a study focused on soil management practices	Binary (0 or 1)	0.41 (0.50)	0.12 (0.33)
MGMT_WATER	Dummy variable indicating a study focused on water management practices	Binary (0 or 1)	0.74 (0.45)	0.76 (0.44)
MGMT_FERT	Dummy variable indicating a study focused on fertilizer management practices	Binary (0 or 1)	0.19 (0.40)	0.20 (0.41)
IWUE	Dummy variable indicating crop irrigation water use efficiency (IWUE) was studied	Binary (0 or 1)	0.44 (0.51)	0.44 (0.51)
MULCH	Dummy variable indicating film or residue mulching was used	Binary (0 or 1)	0.22 (0.42)	0.28 (0.46)
CHN	Dummy variable indicating a study was conducted in China	Binary (0 or 1)	0.56 (0.51)	0.24 (0.44)

[a] S.D. denotes standard deviation.

In this study, the mean values of WUE for wheat and cotton were calculated from each empirical study and used as a dependent variable in the meta-regressions.

3.3. Independent variables

In applying meta-regression analysis, both moderator and predictor variables can be included. As proposed by Stanley et al. (2008), among others, the moderator variables can include: measures of estimate's precision, quality measures of original studies/empirical models, characteristics of the author and data, etc. The predictor variables that are of interest include: adoption of micro irrigation systems, farm management practices on crop, soil, water and fertilizer, etc. All the independent variables and their descriptions and measurements are presented in Table 1.

3.3.1. Measures of estimate's precision

Two measures are identified and used in this meta-analysis. *Se* is the standard error of WUE from each empirical study. *SQRT(n)* is a measure based on the number of observations reported. Both measures are continuous and can be used to account for publication bias and deal with heteroscedasticity (Liu and Shumway, 2016; Stanley et al., 2008).

3.3.2. Characteristics of the data

YEAR represents the year in which the experiment of an empirical study started (converted to an index by subtracting 1980). *NUM_YEAR* refers to the number of years that an empirical study (or an experiment) lasted. *NUM_DATA* is the number of data points obtained from an empirical study. These three measures reflect the characteristics of the data and they are continuous variables. All the rest independent variables are binary.

3.3.3. Adoption of micro irrigation systems

IRRI_MICRO is a dummy variable indicating the application of micro irrigation when conducting the field experiment (1 if applying micro irrigation, and 0 if furrow irrigation).

3.3.4. Climate of the study area

CLIM_ARID indicates an arid or semiarid climate of the study area (1 if arid or semiarid, and 0 otherwise).

3.3.5. Farm management practices

MGMT_CROP, *MGMT_SOIL*, *MGMT_WATER*, and *MGMT_FERT* indicate the empirical study investigated the effects of various farm management practices focusing on crop, soil, water and fertilizer (1 if yes, and 0 if no).

3.3.6. Characteristics of the empirical study[5]

IWUE indicates the study analyzed the crop irrigation water use efficiency (IWUE). *MULCH* means film or residue mulching was adopted in the experiment. *CHN* means the study area was in China. All the three measures are binary (1 if yes, and 0 if no).

The hypotheses are: 1) the moderator variables have mixed effects on crop WUE (Liu and Shumway, 2016; Stanley et al., 2008); and 2) adoption of micro irrigation and other farm management strategies have positive effects on WUE as these include general BMPs (Kang et al., 2012; Sun et al., 2006; Tolk et al., 1999; Xie et al., 2005; Zhao et al., 2012).

[5] The coding of the selected studies originally included several other variables, including precipitation, more categories of climates, and fertilizer application rate. Given the primary goal of this study and multicollinearity concerns, these variables were finally excluded from the analysis to ease the analysis and interpretation of the results.

3.4. Econometric issues

To achieve efficient, consistent, unbiased and robust estimates, multiple assumptions need to be examined before estimating regression models (Florax, 2002; Liu and Shumway, 2016; Nelson and Kennedy, 2009). The empirical studies included in the meta-analysis were distinct and independent. Correlations across observations should not be inferred. However, independence of the cases may be a problem for cotton as one mean WUE value was used in all but three studies that reported WUE using both irrigation methods. In this case, failure to account for correlation in the same study may cause under-estimation of standard errors (Liu and Shumway, 2016). The Lagrange multiplier (LM) test can evaluate this issue and has been used in meta-analysis (Florax et al., 2005). Multicollinearity among independent variables can be a problem for a meta-regression analysis with a relatively small number of studies and many dummy variables (Florax et al., 2005). A standard way to detect the existence of multicollinearity is to calculate the variance inflation factor (VIF) for each independent variable. A VIF less than 10 would indicate no concern with multicollinearity. Heteroskedasticity can also be a problem for meta-regression analysis as the variances for the estimates of WUE in corresponding to each study would likely be unequal. The White test or Breusch-Pagan test can be used to detect this problem. To deal with heteroskedasticity, a WLS approach is commonly employed in meta-regression analysis (Liu and Shumway, 2016; Stanley et al., 2008). Three kinds of weights can be used, including the inverse of the standard error (Stanley et al., 2008), the square root of sample size in each empirical study (Florax et al., 2005; Stern, 2012), and estimated weights from the standard deviation function (Greene, 2005). The specification of WLS using the three weights will be discussed in the model specification section below. In addition, the robustness can be checked by comparing estimates across various models, including ordinary least squares (OLS) and WLS models (Hubert and Rousseeuw, 1997; Tansey et al., 1996). The WLS is also reported to be robust in the presence of potentially omitted variables (Liu and Shumway, 2016).

3.5. Econometric models for meta-regression analysis

This section follows Stanley et al. (2008) to build the econometric models for the *meta*-analysis of crop WUE. A conventional regression model in applied economics research takes the form:

$$Y = X\beta + \varepsilon \tag{4}$$

where Y is the dependent variable (a $n \times 1$ vector, where n is the number of observations), X are the independent variable (a $n \times k$ vector, where k is the number of independent variables), and ε is the random error vector with mean zero and varianceσ^2. $\beta$are the regression coefficients to be estimated (a $k \times 1$ vector).

In this study, the dependent variable is the mean of crop water use efficiency reported in each empirical study, and the independent variables include the predictor variables and moderator variables related to each article as proposed by Stanley and Jarrell (1989) and Stanley et al. (2008). The OLS regression model can be specified as:

$$WUE = \beta + \sum_{k=1}^{K} \alpha_k Z_{jk} + e_j \quad (j = 1, 2 \ldots J) \tag{5}$$

where *WUE* is water use efficiency, and Z_{jk} is a vector containing all independent variables.

Alternatively, following Stanley et al. (2008) the Eq. (5) can be expressed in a generic form to ease the following derivation of the meta-regression models.

$$b_j = \beta + \sum_{k=1}^{K} \alpha_k Z_{jk} + e_j \quad (j = 1, 2 \ldots J) \tag{5'}$$

where b_j represents WUE.

To correct the publication selection bias, according to Stanley et al. (2008), the MRA model can include the standard errors corresponding to the reported mean WUE.

$$b_j = \beta + \sum_{k=1}^{K} \alpha_k Z_{jk} + \beta_0 Se_j + e_j \quad (j = 1, 2 \ldots J) \tag{6}$$

where β is the "true" effect when no publication selection and misspecification biases are present, and Se_j is the standard error of the sample mean WUE b_j in the jth empirical study.

The MRA estimation procedure using OLS may suffer from the heteroskedasticity problem (Stanley and Jarrell, 1989). Given the pooled nature of the meta-data, i.e., separate publications, distinct sample size, differing climate conditions for each study, the OLS regression may likely result in unequal variances. Thus Eq. (6) is also estimated using weighted least squares. As mentioned above, three weights can be used in WLS to obtain efficient estimates.

The WLS model with the inverse of standard errors as weights can be expressed as:

$$b_j^{se} = \beta_0 + \frac{\beta}{Se_j} + \sum_{k=1}^{K} \alpha_k \frac{Z_{jk}}{Se_j} + u_j \quad (j = 1, 2 \ldots J) \tag{7}$$

where $b_j^{se} = \frac{b_j}{Se_j}$ is the weighted WUE estimate, and $u_j = \frac{e_j}{Se_j}$ is the error term with mean zero and variance σ_u^2. β_0 is the new intercept and its testing using t-test can indicate the effect of publication bias, i.e., FAT, and the slope coefficient β can be tested for authentic effect (correction for publication selection bias), i.e., PET (Stanley, 2005).

In MRA, if the standard errors are not provided or calculated in empirical studies, the square root of sample size can also be used to account for publication selection bias, and accordingly they can be alternative weights to deal with heteroskedasticity. The econometric model specified in Eqs. (6) and (7) can be expressed as:

$$b_j = \beta + \sum_{k=1}^{K} \alpha_k Z_{jk} + \beta_0 \frac{1}{\sqrt{n_j}} + e_j \quad (j = 1, 2 \ldots J) \tag{6'}$$

$$b_j^n = \beta_0 + \beta \sqrt{n_j} + \sum_{k=1}^{K} \sqrt{n_j} \alpha_k Z_{jk} + \nu_j \quad (j = 1, 2 \ldots J) \tag{7'}$$

where n_j is the number of observations in the jth empirical study, $b_j^n = \sqrt{n_j} b_j$ is the weighted WUE estimate, and $\nu_j = \sqrt{n_j} e_j$ is the new error term with mean zero and variance σ_ν^2.

Alternative, estimated weights from the standard deviation function related with the OLS model can also be used in WLS models. Following Eq. (6), the WLS model can be expressed as:

$$b_j^{sd} = \beta_0' + \frac{\beta}{\hat{\sigma}_j^2} + \sum_{k=1}^{K} \alpha_k \frac{Z_{jk}}{\hat{\sigma}_j^2} + \omega_j \quad (j = 1, 2 \ldots J) \tag{7"}$$

where $\hat{\sigma}_j^2$ is the estimate of standard deviation σ_j^2, from the regression of squared residuals related with j^{th} empirical study, $b_j^{sd} = \frac{b_j}{\hat{\sigma}_j^2}$

Table 2
Summary of the database for meta-analysis.

		No. of cases	No. of publications	No. of countries	No. of observations
Wheat	Pooled	27	27	9	396
	Furrow	14			205
	Micro irrigation	13			191
Cotton	Pooled [a]	25	22	6	230
	Furrow	13			122
	Micro irrigation	12			108

[a] Three articles (i.e., Cetin and Bilgel (2002), Ibragimov et al. (2007), and Komilov et al. (2003)) reported cotton WUE under both furrow and micro irrigation systems and thus both irrigation systems were included in the meta-analysis.

is the weighted WUE estimate, $\beta_0^{'} = \beta_0 \frac{Se_j}{\hat{\sigma}_j^2}$ is the new intercept, and $\omega_j = \frac{e_j}{\hat{\sigma}_j^2}$ is the new error term with mean zero and variance σ_ω^2.

Using the data on crop WUE from empirical studies, estimation of the three WLS models will be illustrated and compared with the results from OLS models. All the statistical analyses were conducted using the software SAS Version 9.4.

4. Results

4.1. Meta-data overview

An overview of the dataset is presented in Table 2. For wheat, 27 data sources across 9 countries with totally 396 observations were identified. For furrow irrigation, 14 publications with 205 observations were found, and for micro irrigation, 13 publications with 191 observations were selected. Data on cotton were found from 22 sources across 6 countries with totally 230 observations. There were 13 publications using furrow irrigation and 12 cases using micro irrigation, with 122 and 108 observations, respectively.

The meta-data in this study comprise the 27 cases gathered from 27 unique studies on wheat and 25 cases from 22 studies on cotton. All of these studies were published from 1986 to 2012, and 42 out of th total 49 articles were published in 2000 or after. The Table A2 in Appendix A presents data on all the variables obtained from selected publications. For cotton, the number of data sources exceeds the number of publications because three studies, Cetin and Bilgel (2002), Ibragimov et al. (2007) and Komilov et al. (2003), provided data under both furrow and micro irrigation systems. For other publications, varying observations were found under furrow or micro irrigation, thus each publication was treated as an individual case, and the mean values for yield, ET and WUE were calculated from each individual case. Note that the number of cases for wheat and cotton were different and less than the cases considered in some previous meta-analysis (Dalhuisen et al., 2003; Nelson and Kennedy, 2009; Rosenberger and Loomis, 2000; Woodward and Wui, 2001). There are three reasons: (i) as one of the objectives is to determine crop WUE, diverse publications are identified and combined to investigate Yield-ET and WUE-ET relations; (ii) all selected studies should be comparable and must have valid values for all the predictor and moderator variables to facilitate the analysis and integration; and (iii) reduction of stress on the data over expansion of data for meta-analysis should be favored (Bateman and Jones, 2003).

Table 3
Descriptive statistics for yield (kg ha^{-1}), ET(mm) and WUE (kg m^{-3}) of wheat and cotton. The minimum, maximum, mean, median, standard deviation (S.D.) and coefficient of variation (C.V.) of the data are calculated on a pooled basis of the empirical studies.

	Minimum	Maximum	Mean	Median	S.D.	C.V.
Wheat (N = 27)						
Yield	1642.67	6965.13	4482.47	4459.11	1590.03	0.35
ET	229.78	713.10	399.01	374.07	122.33	0.31
WUE	0.54	1.73	1.13	1.15	0.29	0.26
Cotton (N = 25)						
Yield	1345.67	5750.00	3024.12	3057.60	1211.65	0.40
ET	272.90	895.50	547.30	552.20	152.71	0.28
WUE	0.15	1.32	0.60	0.59	0.27	0.46

4.2. Yield-ET relations of wheat and cotton

Table 3 presents summary statistics of yield, ET and WUE with data from empirical studies on a pooled basis. Since all the pooled data are analyzed, this study identified large ranges and moderate variations of yield for both wheat and cotton (CV = 0.35 and 0.40, respectively). The ET range of 272.90–895.50 mm for cotton is broader than wheat's 229.78–713.10 mm. The ranges of WUE for wheat and cotton are 0.54–1.73 kg m^{-3} and 0.15–1.32 kg m^{-3}, with means 1.13 and 0.60 kg m^{-3}, respectively.

It has been reported that the crop yield and ET relations can be presented by parabolic curves for both wheat (Kang et al., 2002; Sun et al., 2006; Wang et al., 2001; Zhang et al., 2006; Zhang and Oweis, 1999; Zhang et al., 2004 and cotton (Cetin and Bilgel, 2002; Howell et al., 2004; Howell et al., 1984; Wanjura et al., 2002). Fig. 2 depicts the yield-ET relations for wheat and cotton under furrow and micro irrigation systems,[6] and Table 4 presents a comparison of scenarios with alternative goals, i.e., max yield *vs.* max WUE. The optimal ET values for wheat are estimated to be 667 mm and 653 mm under furrow and micro irrigation to achieve maximum grain yields of 6531 kg ha^{-1} and 6245 kg ha^{-1}, respectively. It indicates that to achieve the same goal, maximum yield, micro irrigation reduces water use by 2.1% and decreases grain yield by 4.4% compared with the performance under furrow irrigation. The optimal ET values for cotton are estimated to be 604 mm and 467 mm for furrow and micro irrigation to achieve the maximum seed cotton yields of 3221 kg ha^{-1} and 4403 kg ha^{-1}, respectively. This implies that to achieve maximum yield, the water use reduces by 22.7% and yield increases by 36.7% through applying micro irrigation systems

[6] As recommended by one reviewer and due to potential publication bias, the additional figures using all data points from the selected publications are presented in Figs. A1 and A2 in Appendix A.

Table 4
Comparison of scenarios with alternative goals.

	Furrow irrigation			Micro irrigation			Both irrigations		
	Max yield	Max WUE	Δ % [a]	Max yield	Max WUE	Δ %	Max yield	Max WUE	Δ %
Wheat									
Yield	6531	5135	−21.38	6245	4484	−28.20	6325	5389	−14.80
Water use	667	421	−36.79	653	375	−42.57	646	450	−30.37
Cotton									
Yield	3221	2705	−16.03	4403	2778	−36.91	3519	1700	−51.70
Water use	604	450	−25.44	467	273	−41.54	562	273	−51.41

[a] Δ % is percentage change if the scenario changes from maximum yield to maximum WUE.

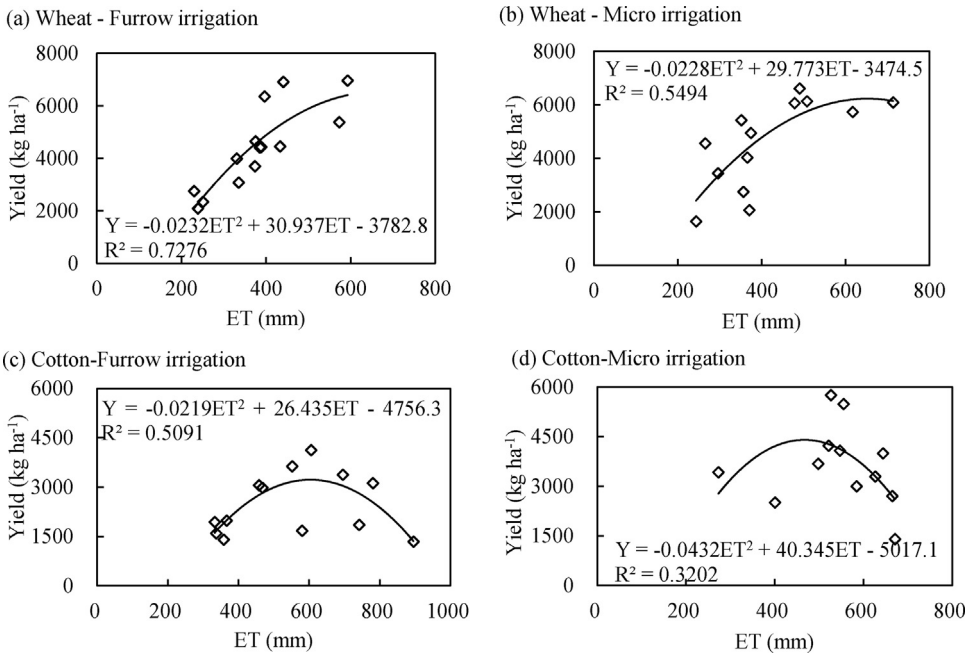


Fig. 2. The yield (Y) and evapotranspiration (ET) relations for wheat and cotton under furrow and micro irrigation systems. For wheat, the grain yield is used to show the yield-ET relation and to calculate WUE, while for cotton, the yield of seed cotton is used.

Table 5
Water saving potentials with 10% yield reduction.

	Furrow irrigation			Micro irrigation			Both irrigations		
	Max yield	After 10% yield reduction	Δ %[a]	Max yield	After 10% yield reduction	Δ %	Max yield	After 10% yield reduction	Δ %
Wheat									
Yield	6531	5878	−10.00	6245	5621	−10.00	6325	5693	−10.00
Water use	667	499	−25.19	653	487	−25.36	646	485	−24.93
Cotton									
Yield	3221	2899	−10.00	4403	3963	−10.00	3519	3167	−10.00
Water use	604	482	−20.15	467	366	−21.62	562	435	−22.63

[a] Δ % is percentage change from maximum yield to after 10% yield reduction.

Table 6
Water saving potentials when compared with water use to achieve maximum yield and maximum WUE.

	Furrow irrigation (%)			Micro irrigation (%)			Both irrigations (%)		
	Mean	S.D.	C.V.	Mean	S.D.	C.V.	Mean	S.D.	C.V.
Maximum yield									
Wheat	14.52	11.95	82.32	2.38	19.21	807.80	8.67	16.73	192.88
Cotton	15.22	16.05	105.46	17.03	16.72	98.15	16.09	16.05	99.78
Maximum WUE									
Wheat	−27.76	42.49	153.05	−13.70	26.78	195.54	−20.99	35.85	170.79
Cotton	−14.59	21.14	144.93	−30.88	42.18	136.57	−22.41	33.28	148.53

S.D. and C.V. denote standard deviation and coefficient of variation. These values are calculated based on means of water use changes related to all data points in the selected publications. All values are presented as percentage to ease interpretation. For the maximum yield option, positive values for means of water use changes show more water is used to achieve maximum yield compared to all other yield levels, indicating potential water saving for other yield levels. For the maximum WUE option, the negative values suggest potential water saving can be achieved to maximize WUE.

instead of furrow irrigation. In addition, under each irrigation system, if the goal changes from maximum yield to maximum WUE, it's estimated that the percentage of water reduction is larger than that of yield reduction. For instance, to maximize wheat WUE under furrow irrigation, about 36.8% of water can be saved with a yield decrease of 21.4%, compared to maximizing yield.

4.3. Water saving potentials

In the previous section, both yield and water use changes are under extreme cases by comparing max yield and max WUE. The water saving potentials can be better understood under the scenario of a 10% yield reduction, and by comparing all water input levels to achieve maximum yield and maximum WUE. As presented in Table 5, for a 10% yield reduction, wheat water use can decrease by 25%, and cotton water use can reduce by 20% and 22% for furrow and micro irrigation, respectively.

The results of water saving under maximum yield and maximum WUE are presented in Table 6. Positive numbers for means of water changes indicates more water is applied to achieve maximum yield, while negative numbers with maximum WUE scenario suggest direct water saving. Furrow irrigation uses 15% more water to

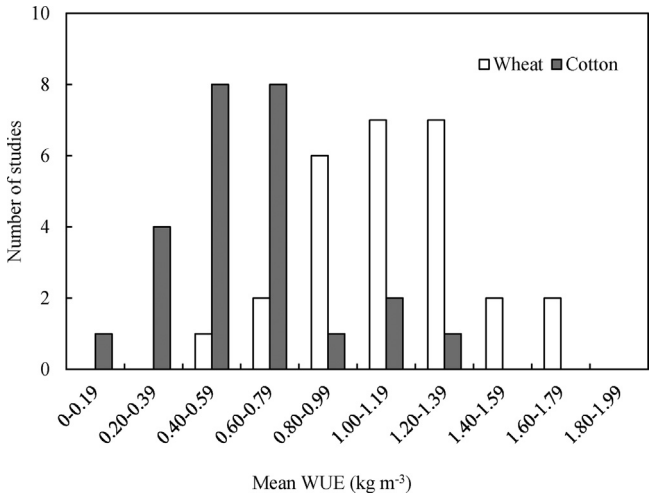


Fig. 3. Histogram of mean WUE of wheat and cotton based on the pooled data of both furrow and micro irrigations.

achieve maximum wheat yield, and micro irrigation uses 2% more. For cotton, about 15% and 17% more water is applied to get maximum yields. This indicates potential water saving if water can be cut off to achieve sub-optimal yield levels. Under the maximum WUE scenario, about 28% and 14% less water is used under furrow and micro irrigation on wheat farms compared with water uses for other WUE levels. To achieve maximum WUE of cotton, about 15% and 31% of water can be saved.

4.4. WUE of wheat and cotton

Fig. 3 presents a histogram of wheat and cotton WUE values with patterns of their distributions.[7] For the 27 cases of wheat, 7 studies present a mean WUE in the middle category, i.e., 1.00–1.19 kg m^{-3}. Two categories (1.20–1.39 and 1.60–1.79) on the right side have one more case than the two (0.40–0.59 and 0.80–0.99) on the left side, respectively. For the 25 cases of cotton, 8 cases have a mean WUE between 0.40 and 0.59 kg m^{-3}, 8 cases between 0.60 and 0.79 kg m^{-3}, and only 1 case between 0.80 and 0.99 kg m^{-3}. The lower category (i.e., 0.20–0.39) has two more cases than the upper category (i.e., 1.00–1.19). Nevertheless, the fairly symmetric histograms indicate the WUE values for wheat and cotton might not be normally distributed.

Quadratic relations between crop WUE and ET have been identified for wheat (Kang et al., 2002; Sezen and Yazar, 2006) and cotton (Howell et al., 2004) through conducting regression analysis using field experiment data. Fig. 4 indicates the relationships between WUE and ET of wheat and cotton with the pooled data under both furrow and micro irrigation systems.[8] The optimal ET for wheat and cotton are 450 mm and 273 mm to realize the estimated maximum WUE, 1.24 kg m^{-3} and 1.32 kg m^{-3}, respectively. While through analyzing yield-ET relations of the pooled data, the optimal ET is 646 mm to achieve the maximum yield of 6325 kg ha^{-1} for wheat, and 562 mm to achieve the maximum yield of 3519 kg ha^{-1} for cotton (Table 4). This finding implies that the maximum WUE can be realized by reducing crop water use by 30.4% and 51.4% for wheat and cotton, respectively, compared with the amount of irrigation water to achieve the maximum yields. Meanwhile, yields decrease by 14.8% for wheat and by 51.7% for cotton. This confirms the water

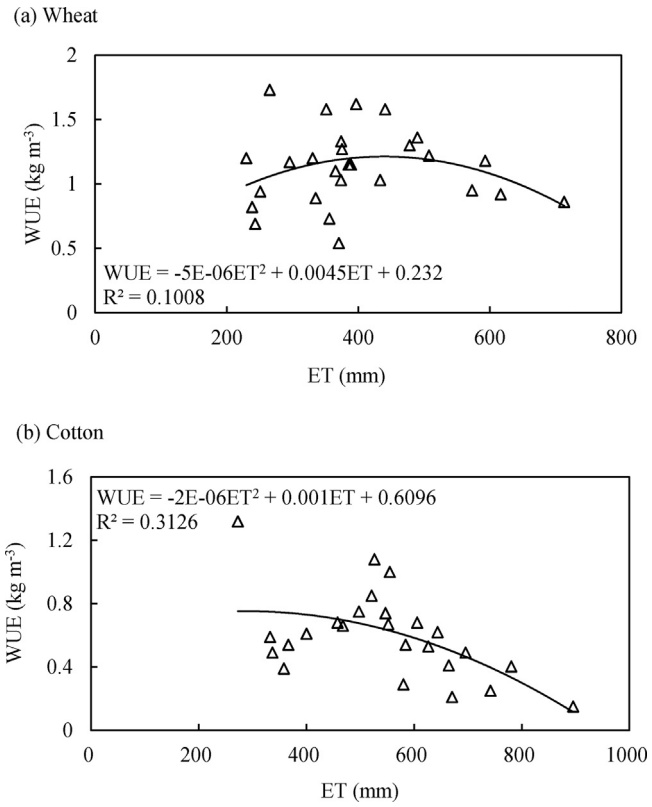


Fig. 4. The WUE-ET relations using the pooled data of (a) wheat and (b) cotton.

saving potentials presented earlier. As recommendation by many researchers, crop WUE could be significantly improved with reduction of irrigation water, for instance, using deficit irrigation (Jalota et al., 2006; Unlü et al., 2011; Zhang et al., 2004) and partial root-zone irrigation/drying (Du et al., 2006; Tang et al., 2010), etc.

4.5. Results of meta-regression analysis

To visually check for the publication selection bias, the funnel plots are presented in Fig. 5, which portrays the WUE values with the precision of their estimates, 1/Se$_j$. The scatterplot of WUE values should form an inverted funnel if no publication selection bias is present (Stanley, 2005). However, the two funnel graphs look fairly like funnels, which may indicate a publication selection bias or not. (Alternative funnel plots using square roots of sample size cannot really conclude with possible existence of pubilicaiton bias either, see Fig. A5 in Appendix A.) In addition, the funnel asymmetry test can be conducted (Tables 7 and 8). Four models are estimated for each crop, including OLS (Model 1), and WLS models (i.e., Model 2 with the inverse of standard errors as weights, Model 3 with square root of sample size as weights, and Model 4 with estimated weights from the standard deviation function).[9]

The *t*-tests of the intercepts in the regression models indicate the null hypothsis of the absence of publication bias is rejected for wheat (Models 1, 2, and 4), while retained for cotton. Hence, significant publication bias are found in wheat WUE. To test for genuine effects beyond the publication bias, *t*-tests of the slopes ($1/Se_j$, $\sqrt{n_j}$ and $1/\hat{\sigma}_j^2$) in WLS models can be used. Significant precision-effect tests show clear evidence of WUE after the publication selection is corrected ($p < 0.01$ in Model 4 of wheat; $p < 0.10$ in Model 2, and

[7] Recommended by one reviewer, additional histograms plotted using all data points are presented in Fig. A3 in Appendix A.

[8] As recommended by one reviewer, additional figures using all data points are presented in Fig. A4 in Appendix A.

[9] As recommended by one reviewer, values of AIC for all the models are presented in Table A3 in Appendix A.

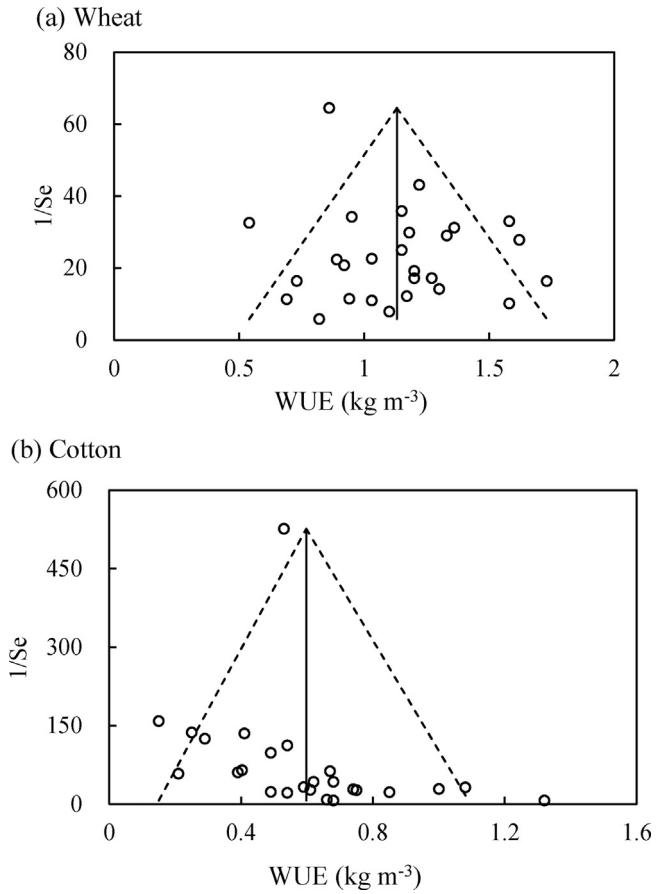


Fig. 5. Funnel plots of WUE using standard errors of (a) wheat and (b) cotton.

p < 0.01 in Model 4 of cotton). A positive coefficient suggests studies with higher WUE are more likely to be published or accpeted for publication by editors, while a negative coefficient indicates studies with lowerer WUE are more likely to be published.

The empirical estimates of the meta-regression models for crop water use efficiency using OLS and WLS are presented in Table 7 for wheat and Table 8 for cotton. Wheat regressions are estimated using 27 cases, and cotton regressions estimated using 25 cases. The adjusted R^2 values are 0.439-0.987 for wheat models, and 0.858-0.962 for cotton models, which are close to the adjusted R^2 values in previous meta-regression analysis by Liu and Shumway (2016), Nelson and Kennedy (2009), and Stanley et al. (2008). The mean VIF values are 2.53 and 2.90 for wheat and cotton equations, respecively, which indicates no concern with multicollinearity.

Overall, more independent variables are significant in predicting cotton WUE than in predicting wheat WUE. The following interpretation will focus on the WLS models as moderate heteroskedasticity exists in both wheat and cotton OLS models (0.05 < p < 0.10). Regarding the characteristics of data, coefficient of the year variable is positive and statistically significant in wheat models, indicating more recent studies reported higher WUE. The estimate of the number of years for a study is negative for cotton (Model 4), and this suggests studies reporting results of short-time experiments tended to report higher WUE. Furthermore, the number of data is statistically significant, and the effect is negative for wheat (Model 4) and positive for cotton (Models 2 and 4). While the negative effect on wheat WUE is unexpected, these findings on cotton indicate more data presented in an empirical study are likely to be associated with higher cotton WUE. This is probably because more data signal to editors that more work has been done, and thus the article is more likely to be accepted. One effect of publication bias is to tempt authors to search for larger estimates as large-sample studies tend to have small standard errors (Stanley et al., 2008).

Table 7
Meta-regression analysis for water use efficiency (WUE) of *wheat* using ordinary least squares (OLS) and weighted least squares (WLS) models.

		WLS		
Variable	Model 1: OLS Coefficient	Model 2 Coefficient	Model 3 Coefficient	Model 4 Coefficient
Intercept	0.620**	0.487*	0.985	0.710***
	(0.288)	(0.249)	(0.573)	(0.124)
Publication bias Correction				
Se	−2.183			−2.907***
	(2.207)			(0.971)
1/Se		−0.001		
		(0.005)		
SQRT(n)			−0.425	
			(0.362)	
YEAR	0.016	0.018*	0.021**	0.015**
	(0.011)	(0.009)	(0.010)	(0.006)
NUM_YEAR	0.022	0.027	0.033	0.010
	(0.026)	(0.020)	(0.022)	(0.011)
NUM_DATA	−0.010	−0.005	0.050	−0.010**
	(0.007)	(0.008)	(0.046)	(0.004)
IRRI_MICRO	0.180	0.212*	0.183	0.159**
	(0.133)	(0.118)	(0.120)	(0.066)
CLIM_ARID	−0.302*	−0.367**	−0.341**	−0.233**
	(0.153)	(0.138)	(0.131)	(0.085)
MGMT_CROP	−0.248*	−0.205*	−0.221*	−0.190**
	(0.123)	(0.111)	(0.114)	(0.072)
MGMT_SOIL	0.414**	0.408***	0.414***	0.432***
	(0.158)	(0.135)	(0.134)	(0.058)
MGMT_WATER	0.454**	0.413**	0.535***	0.498***
	(0.168)	(0.152)	(0.168)	(0.078)
MGMT_FERT	−0.021	−0.170	−0.003	−0.049
	(0.143)	(0.126)	(0.125)	(0.090)
IWUE	0.022			
	(0.139)			
MULCH	0.116	0.102	0.225	0.040
	(0.171)	(0.121)	(0.131)	(0.031)
CHN	0.032			
	(0.109)			
Model Fit				
N	27	27	27	27
R-Sq	0.719	0.739	0.727	0.992
Adj R-Sq	0.439	0.547	0.528	0.987
F	2.560	3.860	3.640	179.220
Pr > F	0.051	0.009	0.011	<0.001
Mean VIF	2.530			

Standard errors are in parentheses.
*** Significant at 1% level.
** Significant at 5% level.
* Significant at 10% level.

The coefficient of the dummy variable, micro irrigation, is positive and statistically significant in both wheat and cotton models but at different significant levels. In line with the expectation, this indicates the adoption of micro irrigation can increase WUE through reducing both the amount of irrigation water and water for evaporation (Ibragimov et al., 2007; Wanjura et al., 2002). The dummy variable on arid or semiarid climate is significant in predicting WUE, and its effect is negative for wheat and positive for cotton. This difference may be because cotton tends to tolerate dry conditions better than wheat. Water-consuming crops in arid or semiarid climate may require more water as a large proportion of the water can evaporate, while drought-tolerant crops can perform better with less irrigation. If scheduled well and complemented with appropriate BMPs, however, water can be saved in arid areas compared with the crop water consumption in humid climates.

The dummy variable regarding the adoption of crop management practices shows a negative effect on wheat, while a positive effect on cotton. Such practices like better varieties and double

Table 8
Meta-regression analysis for water use efficiency (WUE) of *cotton* using ordinary least squares (OLS) and weighted least squares (WLS) models.

		WLS		
Variable	OLS: Model 1 Estimate	Model 2 Coefficient	Model 3 Coefficient	Model 4 Coefficient
Intercept	−0.014	−0.020	−0.015	0.035
	(0.133)	(0.091)	(0.294)	(0.088)
Publication bias Correction				
Se	1.425***			1.326***
	(0.616)			(0.376)
1/Se		−0.00024*		
		(0.00012)		
SQRT(n)			0.093	
			(0.181)	
YEAR	0.006	0.006	0.002	0.003
	(0.006)	(0.005)	(0.006)	(0.004)
NUM_YEAR	−0.006	−0.009	−0.016	−0.015*
	(0.012)	(0.010)	(0.012)	(0.008)
NUM_DATA	0.007	0.013**	−0.006	0.011**
	(0.006)	(0.005)	(0.028)	(0.004)
IRRI_MICRO	0.108**	0.083*	0.093*	0.102***
	(0.046)	(0.042)	(0.053)	(0.027)
CLIM_ARID	0.209**	0.258***	0.200**	0.247***
	(0.068)	(0.049)	(0.076)	(0.046)
MGMT_CROP	0.143**	0.173***	0.215***	0.117***
	(0.061)	(0.052)	(0.067)	(0.036)
MGMT_SOIL	0.305**	0.341***	0.356***	0.402***
	(0.105)	(0.114)	(0.116)	(0.097)
MGMT_WATER	0.069	0.085*	0.080	0.063
	(0.063)	(0.048)	(0.072)	(0.042)
MGMT_FERT	−0.004	−0.014	−0.013	0.005
	(0.057)	(0.046)	(0.068)	(0.042)
IWUE	0.114**	0.090*	0.148**	0.130***
	(0.052)	(0.049)	(0.054)	(0.040)
MULCH	−0.034	−0.078	−0.049	−0.075
	(0.087)	(0.068)	(0.086)	(0.066)
CHN	0.010			
	(0.062)			
Model Fit				
N	25	25	25	25
R-Sq	0.951	0.952	0.929	0.981
Adj R-Sq	0.893	0.903	0.858	0.962
F Value	16.410	19.610	13.090	50.970
Pr > F	<0.001	<0.001	<0.001	<0.001
Mean VIF	2.900			

Standard errors are in parentheses.
*** Significant at 1% level.
** Significant at 5% level.
* Significant at 10% level.

cropping can reduce irrigation water use or improve yield, thus resulting in higher cotton WUE (Ingram et al., 2012). However, the negative effect in wheat models is out of expectation, but very interesting, and more empirical work is needed to investigate the reasons. The effect of soil management practices is positive and significant for both wheat and cotton. This finding suggests soil practices are most effective in improving WUE of both wheat and cotton. These practices, for instance, alternative/rotational tillage, and terracing maintenance, can be applicable for various crops, and thus improved soil condition can prevent over-irrigation and increase crop yield (Ingram et al., 2012; Jalota et al., 2008). The variable regarding the application of water management practices is significant for wheat and cotton (Model 2 only). Consistent with the findings from many empirical studies, conservation-oriented water practices can contribute to higher WUE by determining when and how much water is needed through investigating crop water requirements and scientifically scheduling irrigation (Cayci et al., 2009; Kang et al., 2012). However the fertilizer management practices are not significant in predicting WUE.

The results show investigating IWUE improves cotton WUE. Different from WUE, IWUE measures the crop water use efficiency while excluding the allowance of rainfall in the growing season. As IWUE is always greater than WUE for the same crop growing in the same season (Jiang et al., 2012; Sezen and Yazar, 2006), authors investigating both measures in one study are tempted to matching WUE with IWUE, resulting in higher WUE values which can be favored in publication. The variables, field mulching and studies conducted in China, are not significant.

Robustness checks can be conducted through comparing the changes of signs and significance levels of the coefficients across multiple models. For wheat models, the signs of coefficients mostly remain consistent, while three variables, YEAR, NUM_DATA, and IRRI_MICRO, are only significant in the WLS models. Similarly, the outcomes from cotton models show consistent signs and significant levels, while NUM_YEAR, NUM_DATA, and MGMT_WATER are not significant in the OLS model. This indicates the advantage of WLS to deal with heteroskedasticity and provide consistent and reliable estimates in meta-regression analysis.

5. Concluding remarks

In this research, a meta-regression analysis of crop WUE are conducted focusing on 52 cases from 49 empirical studies published in 1986–2012. Through extracting data in empirical publications, a detailed database was obtained on wheat and cotton including mean water use, yield, WUE, irrigation systems employed, multiple farm management practices, as well as other moderator variables related to the publications and study areas. With the unique dataset, this study tries to determine a comparable base of wheat and cotton WUE, explore yield-ET relations, and evaluate the effects of micro irrigation and other farm BMPs on improving WUE. In the course of this exercise, several interesting findings are uncovered.

1) Water use is different under different irrigation systems and also varying for different crops. Comparisons of results based on the pooled data under two irrigation systems suggest that to achieve maximum yield of wheat under micro irrigation, water use reduces by 2.1% and grain yield decreases by 4.4%, compared with furrow irrigation. To achieve maximum yield of cotton under micro irrigation, a 22.7% reduction in water use increases the yield by 37.7%, compared with furrow irrigation.
2) Broad benchmarks of WUE are identified for wheat and cotton, i.e., 0.54–1.73 kg m^{-3} and 0.15–1.32 kg m^{-3}, respectively, and the mean values are 1.13 and 0.60 kg m^{-3}.
3) Based on estimated quadratic relations between WUE and ET, 30.4% and 51.4% of water use can be saved for wheat and cotton if the goal of agricultural production is to achieve maximum WUE rather than maximum yields. As a result, crop yields decrease by 14.8% for wheat and 51.7% for cotton.
4) Alternative scenarios show an exciting potential of water saving. For a 10% yield reduction, water use can decrease by 25% in wheat fields, and by 20–22% in cotton fields. Compared with water use for maximum yield, other yield levels reduce water use by 2–15% on wheat and by 15–17% on cotton. Under the maximum WUE scenario, wheat water use decreases by 14–28% and cotton water use reduces by 15–31%.

Varying estimates of water use, yield and WUE are probably because the empirical studies were conducted under diverse climate conditions, irrigation systems, farm management practices, and so forth. Variations of these factors need to be considered jointly to evaluate crop WUE integrated from empirical studies and provide insights on options to increase WUE.

To reveal systematic causal patterns for the investigated measures obtained from previous publications, many researchers in applied economics utilized meta-analytical techniques (Liu and Shumway, 2016; Rosenberger and Loomis, 2000; Woodward and Wui, 2001). In this case, meta-regression analyses on wheat and cotton employ one OLS model and three WLS models to explore some fundamental effects of the predictor and moderator variables on a pooled basis. The illustration of MRA models on wheat and cotton WUE indicates distinct and fundamental effects of irrigation systems, management strategies, and attributes of publications, which are consistent with the findings from previous meta-analysis (Johnston et al., 2003). The significant effects from multiple independent variables constitute the general patterns that are expected. For wheat, higher WUE is associated with growing in humid environments, under micro irrigation systems, less crop management practices, and more soil and water management practices. Meanwhile, more practices are effective to increase cotton WUE, including more crop and soil management practices (Levidow et al., 2014) and investigation of IWUE. Some data and publication characteristics like recent experimental results and a larger number of data presented in a study are also associated with higher WUE. The WLS models with appropriate weights not only provide efficient and robust estimates (Liu and Shumway, 2016), but facilitate the testing of publication bias and genuine effects in meta-regression analyses (Stanley, 2005; Stanley et al., 2008).

Although the authors notice some research reporting a potential interaction effect of two or more jointly adopted farm management practices, unfortunately this study cannot test this in the meta-analysis due to limitation of selected empirical studies. An update of the database can be completed later, and more comparable predictor and moderator variables can be selected to investigate adoption of specific BMPs in detail. Furthermore, as mentioned by a reviewer, a direct comparison of micro *vs.* furrow irrigation can be possible through calculating effect size in future research.

Acknowledgements

This research was funded by the National Basic Research Program (973) of China (2014CB138702) and the State Key Laboratory for Grassland Agro-ecosystems at Lanzhou University under project # SKLGAE201502. The authors are very grateful to Zeng Tang and Jinfeng Zhang for their assistance in statistical analysis and comments on an earlier version. The authors are also grateful to the participants at the 2016 Annual Meetings of Agricultural and Applied Economics Association (AAEA). The authors appreciate the constructive comments from the journal editors, Enrique Fernández and Luca Sebastiani, and two anonymous reviewers. The study was conducted when Yubing Fan was a master student at Lanzhou University.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at https://doi.org/10.1016/j.agwat.2017.12.006.

References

Albrizio, R., Todorovic, M., Matic, T., Stellacci, A.M., 2010. Comparing the interactive effects of water and nitrogen on durum wheat and barley grown in a Mediterranean environment. Field Crop. Res. 115, 179–190.
Bateman, I.J., Jones, A.P., 2003. Contrasting conventional with multi-level modeling approaches to meta-analysis: expectation consistency in UK woodland recreation values. Land Econ. 79, 235–258.
Baumhardt, R.L., Jones, O.R., 2002. Residue management and tillage effects on soil-water storage and grain yield of dryland wheat and sorghum for a clay loam in Texas. Soil Till. Res. 68, 71–82.
Bergstrom, J.C., Taylor, L.O., 2006. Using meta-analysis for benefits transfer: theory and practice. Ecol. Econ. 60, 351–360.
Bordovsky, D.G., Choudhary, M., Gerard, C.J., 1999. Effect of tillage, cropping, and residue management on soil properties in the Texas Rolling Plains. Soil Sci. 164, 331–340.
Boyle, K.J., Poe, G.L., Bergstrom, J.C., 1994. What do we know about groundwater values? Preliminary implications from a meta analysis of contingent-valuation studies. Am. J. Agric. Econ. 76, 1055–1061.
Bravo-Ureta, B.E., Solís, D., Moreira López, V.H., Maripani, J.F., Thiam, A., Rivas, T., 2007. Technical efficiency in farming: a meta-regression analysis. J. Productivity Anal. 27, 57–72.
Bucks, D.A., Allen, S.G., Roth, R.L., Gardner, B.R., 1988. Short staple cotton under micro and level-basin irrigation methods. Irrig. Sci. 9, 161–176.
Cayci, G., Heng, L.K., Oztürk, H.S., Sürek, D., Kütük, C., Sağlam, M., 2009. Crop yield and water use efficiency in semi-arid region of Turkey. Soil Till. Res. 103, 65–72.
Cetin, O., Bilgel, L., 2002. Effects of different irrigation methods on shedding and yield of cotton. Agric. Water. Manage. 54, 1–15.
Chakraborty, D., Nagarajan, S., Aggarwal, P., Gupta, V.K., Tomar, R.K., Garg, R.N., Sahoo, R.N., Sarkar, A., Chopra, U.K., Sarma, K.S.S., Kalra, N., 2008. Effect of mulching on soil and plant water status, and the growth and yield of wheat (*Triticum aestivum* L.) in a semi-arid environment. Agric. Water. Manage. 95, 1323–1334.
Dalhuisen, J.M., Florax, R.J.G.M., de Groot, H.L.F., Nijkamp, P., 2003. Price and income elasticities of residential water demand: a meta-analysis. Land Econ. 79, 292–308.
Du, T., Kang, S., Zhang, J., Li, F., Hu, X., 2006. Yield and physiological responses of cotton to partial root-zone irrigation in the oasis field of northwest China. Agric. Water. Manage. 84, 41–52.
Fan, Y., Wang, C., Nan, Z., 2014. Comparative evaluation of crop water use efficiency, economic analysis and net household profit simulation in arid Northwest China. Agric. Water. Manage. 146, 335–345.
Florax, R.J., Travisi, C.M., Nijkamp, P., 2005. A meta-analysis of the willingness to pay for reductions in pesticide risk exposure. Eur. Rev. Agric. Econ. 32, 441–467.
Florax, R.J.G.M., 2002. Accounting for Dependence Among Study Results in Meta-Analysis: Methodology and Applications to the Valuation and Use of Natural Resources, http://degree.ubvu.vu.nl/repec/vua/wpaper/pdf/20020005.pdf. (Accessed September 2016).
Gärdenäs, A.I., Hopmans, J.W., Hanson, B.R., Simůnek, J., 2005. Two-dimensional modeling of nitrate leaching for various fertigation scenarios under micro-irrigation. Agric. Water. Manage. 74, 219–242.
Greene, W.H., 2005. Econometric Analysis, 5th ed. Prentice Hall, New York.
Hodgson, A.S., Constable, G.A., Duddy, G.R., Daniells, I.G., 1990. A comparison of drip and furrow irrigated cotton on a cracking clay soil 2. Water use efficiency, waterlogging, root distribution and soil structure. Irrig. Sci. 11, 143–148.
Hou, X., Li, R., Jia, Z., Han, Q., Wang, W., Yang, B., 2012. Effects of rotational tillage practices on soil properties: winter wheat yields and water-use efficiency in semi-arid areas of north-west China. Field Crop. Res. 129, 7–13.
Howell, T.A., Davis, K.R., McCormick, R.L., Yamada, H., Walhood, V.T., Meek, D.W., 1984. Water use efficiency of narrow row cotton. Irrig. Sci. 5, 195–214.
Howell, T.A., Asce, M., Evett, S.R., Tolk, J.A., Schneider, A.D., 2004. Evapotranspiration of full-, deficit-irrigated, and dryland cotton on the Northern Texas High Plains. J. Irrig. Drain. Eng. 130, 277–285.
Huang, G., Chai, Q., Feng, F., Yu, A., 2012. Effects of different tillage systems on soil properties, root growth, grain yield, and water use efficiency of winter wheat (*Triticum aestivum* L.) in arid Northwest China. J. Integr. Agric. 11, 1286–1296.
Hubert, M., Rousseeuw, P.J., 1997. Robust regression with both continuous and binary regressors. J. Stat. Plann. Inference 57, 153–163.
Ibragimov, N., Evett, S.R., Esanbekov, Y., Kamilov, B.S., Mirzaev, L., Lamers, J.P.A., 2007. Water use efficiency of irrigated cotton in Uzbekistan under drip and furrow irrigation. Agric. Water. Manage. 90, 112–120.
Ingram, J., Mills, J., Frelih-Larsen, A., Davis, M., 2012. Uptake of Soil Management Practices and Experiences with Decisions Support Tools: Analysis of the Consultation with the Farming Community (Project No 289684). Smart SOIL.
Jalota, S.K., Sood, A., Chahal, G.B.S., Choudhury, B.U., 2006. Crop water productivity of cotton (*Gossypium hirsutum* L.)-wheat (*Triticum aestivum* L.) system as influenced by deficit irrigation, soil texture and precipitation. Agric. Water. Manage. 84, 137–146.
Jalota, S.K., Buttar, G.S., Sood, A., Chahal, G.B.S., Ray, S.S., Panigrahy, S., 2008. Effects of sowing date, tillage and residue management on productivity of cotton (*Gossypium hirsutum* L.)-wheat (*Triticum aestivum* L.) system in northwest India. Soil Till. Res. 99, 76–83.
Jiang, J., Huo, Z., Feng, S., Zhang, C., 2012. Effect of irrigation amount and water salinity on water consumption and water productivity of spring wheat in Northwest China. Field Crop. Res. 137, 78–88.
Jin, M., Zhang, R., Sun, L., Gao, Y., 1999. Temporal and spatial soil water management: a case study in the Heilonggang region, PR China. Agric. Water. Manage. 42, 173–187.
Johnston, R.J., Besedin, E.Y., Wardwell, R.F., 2003. Modeling relationships between use and nonuse values for surface water quality: a meta-analysis. Water Resour. Res. 39, 1363.
Kang, S., Zhang, L., Liang, Y., Hu, X., Cai, H., Gu, B., 2002. Effects of limited irrigation on yield and water use efficiency of winter wheat in the Loess Plateau of China. Agric. Water. Manage. 55, 203–216.
Kang, Y., Wang, R., Wan, S., Hu, W., Jiang, S., Liu, S., 2012. Effects of different water levels on cotton growth and water use through drip irrigation in an arid region with saline ground water of Northwest China. Agric. Water. Manage. 109, 117–126.

Komilov, B., Ibragimov, N., Esanbekov, Y., Evett, S.R., Lee, H., 2003. Irrigation scheduling study of drip irrigated cotton by use of soil moisture Neutron probe. In: Proceedings of the UNCGRI/IAEA National Workshop in Optimization of Water and Fertilizer Use for Major Crops of Cotton Rotation, Tashkent, Uzbekistan.
Levidow, L., Zaccaria, D., Maia, R., Vivas, E., Todorovic, M., Scardigno, A., 2014. Improving water-efficient irrigation: prospects and difficulties of innovative practices. Agric. Water. Manage. 146, 84–94.
Li, J., Inanaga, S., Li, Z., Eneji, A.E., 2005. Optimizing irrigation scheduling for winter wheat in the North China Plain. Agric. Water. Manage. 76, 8–23.
List, J.A., Gallet, C.A., 2001. What experimental protocol influence disparities between actual and hypothetical stated values? Environ. Resour. Econ. 20, 241–254.
Liu, B., Shumway, C.R., 2016. Substitution elasticities between GHG-polluting and nonpolluting inputs in agricultural production: a meta-regression. Energy Econ. 54, 123–132.
Liu, W., Hunsaker, D.J., Li, Y., Xie, X., Wall, G.W., 2002. Interrelations of yield, evapotranspiration, and water use efficiency from marginal analysis of water production functions. Agric. Water. Manage. 56, 143–151.
Liu, H., Yu, L., Luo, Y., Wang, X., Huang, G., 2011. Responses of winter wheat (*Triticum aestivum* L.) evapotranspiration and yield to sprinkler irrigation regimes. Agric. Water. Manage. 98, 483–492.
Loomis, J.B., White, D.S., 1996. Economic benefits of rare and endangered species: summary and meta-analysis. Ecol. Econ. 18, 197–206.
Luo, Z., Wang, E., Sun, O.J., 2010. Can no-tillage stimulate carbon sequestration in agricultural soils? A meta-analysis of paired experiments. Agric. Ecosyst. Environ. 139, 224–231.
Mateos, L., Berengena, J., Orgaz, F., Diz, J., Fereres, E., 1991. A comparison between drip and furrow irrigation in cotton at two levels of water supply. Agric. Water. Manage. 19, 313–324.
Murray, S.J., Foster, P.N., Prentice, I.C., 2012. Future global water resources with respect to climate change and water withdrawals as estimated by a dynamic global vegetation model. J. Hydrol. 448–449, 14–29.
Nelson, J.P., Kennedy, P.E., 2009. The use (and abuse) of meta-analysis in environmental and natural resource economics: an assessment. Environ. Resour. Econ. 42, 345–377.
Oweis, T.Y., Farahani, H.J., Hachum, A.Y., 2011. Evapotranspiration and water use of full and deficit irrigated cotton in the Mediterranean environment in northern Syria. Agric. Water. Manage. 98, 1239–1248.
Pagdee, A., Kim, Y.S., Daugherty, P.J., 2006. What makes community forest management successful: a meta-study from community forests throughout the world. Soc. Nat. Resour. 19, 33–52.
Phillips, J.M., 1994. Farmer education and farmer efficiency: a meta-analysis. Econ. Dev. Cult. Change 43, 149–165.
Pittelkow, C.M., Linquist, B.A., Lundy, M.E., Liang, X., van Groenigen, K.J., Lee, J., van Gestel, N., Six, J., Venterea, R.T., van Kessel, C., 2015. When does no-till yield more? A global meta-analysis. Field Crop. Res. 183, 156–168.
Qin, W., Hu, C., Oenema, O., 2015. Soil mulching significantly enhances yields and water and nitrogen use efficiencies of maize and wheat: a meta-analysis. Sci. Rep. 5, 162100.
Qin, W., Assinck, F.B.T., Heinen, M., Oenema, O., 2016. Water and nitrogen use efficiencies in citrus production: a meta-analysis. Agric. Ecosyst. Environ. 222, 103–111.
Roberts, C.J., 2005. Issues in meta-regression analysis: an overview. J. Econ. Surv. 19, 295–298.
Rosenberger, R.S., Loomis, J.B., 2000. Panel stratification in meta-analysis of economic studies: an investigation of its effects in the recreation valuation literature. J. Agr. Appl. Econ. 32, 459–470.
Rosenberger, R.S., Stanley, T.D., 2006. Measurement, generalization, and publication: sources of error in benefit transfers and their management. Ecol. Econ. 60, 372–378.
Sadras, V.O., 2009. Does partial root-zone drying improve irrigation water productivity in the field? A meta-analysis. Irrig. Sci. 27, 183–190.
Santos, J.M.L., 1998. The Economic Valuation of Landscape Change: Theory and Policies for Land Use and Conservation. Edward Elgar Publishing.
Schneider, A.D., Howell, T.A., 2001. Scheduling deficit wheat irrigation with data from an evapotranspiration network. Trans. ASAE 44, 1617–1623.
Sezen, S.M., Yazar, A., 2006. Wheat yield response to line-source sprinkler irrigation in the arid Southeast Anatolia region of Turkey. Agric. Water. Manage. 81, 59–76.
Singh, Y., Rao, S.S., Regar, P.L., 2010. Deficit irrigation and nitrogen effects on seed cotton yield, water productivity and yield response factor in shallow soils of semi-arid environment. Agric. Water. Manage. 97, 965–970.
Smith, V.K., Kaoru, Y., 1990. Signals or noise? Explaining the variation in recreation benefit estimates. Am. J. Agric. Econ. 72, 419–433.
Stanley, T.D., Jarrell, S.B., 1989. Meta-regression analysis: a quantitative method of literature surveys. J. Econ. Surv. 3, 161–170.
Stanley, T.D., Doucouliagos, C., Jarrell, S.B., 2008. Meta-regression analysis as the socio-economics of economics research. J. Soc. Econ. 37, 276–292.
Stanley, T.D., 2001. Wheat from chaff: meta-analysis as quantitative literature review. J. Econ. Perspect. 15, 131–150.
Stanley, T.D., 2005. Beyond publication bias. J. Econ. Surv. 19, 309–345.
Stern, D.I., 2012. Interfuel substitution: a meta-analysis. J. Econ. Surv. 26, 307–331.
Sun, H., Liu, C., Zhang, X., Shen, Y., Zhang, Y., 2006. Effects of irrigation on water balance, yield and WUE of winter wheat in the North China Plain. Agric. Water. Manage. 85, 211–218.
Tang, L., Li, Y., Zhang, J., 2010. Partial rootzone irrigation increases water use efficiency, maintains yield and enhances economic profit of cotton in arid area. Agric. Water. Manage. 97, 1527–1533.
Tansey, R., White, M., Long, R.G., Smith, M., 1996. A comparison of loglinear modeling and logistic regression in management research. J. Manage. 22, 339–358.
Tolk, J.A., Howell, T.A., Evett, S.R., 1999. Effect of mulch, irrigation, and soil type on water use and yield of maize. Soil Till. Res. 50, 137–147.
Tong, L., Kang, S., Zhang, L., 2007. Temporal and spatial variations of evapotranspiration for spring wheat in the Shiyang river basin in northwest China. Agric. Water. Manage. 87, 241–250.
Unlü, M., Kanber, R., Koç, D.L., Tekin, S., Kapur, B., 2011. Effects of deficit irrigation on the yield and yield components of drip irrigated cotton in a Mediterranean environment. Agric. Water. Manage. 98, 597–605.
Van Houtven, G., Powers, J., Pattanayak, S.K., 2007. Valuing water quality improvements in the United States using meta-analysis: is the glass half-full or half-empty for national policy analysis? Resour. Energy Econ. 29, 206–228.
Wang, H., Zhang, L., Dawes, W.R., Liu, C., 2001. Improving water use efficiency of irrigated crops in the North China Plain-measurements and modelling. Agric. Water. Manage. 48, 151–167.
Wanjura, D.F., Upchurch, D.R., Mahan, J.R., Burke, J.J., 2002. Cotton yield and applied water relationships under drip irrigation. Agric. Water. Manage. 55, 217–237.
Woodward, R.T., Wui, Y.S., 2001. The economic value of wetland services: a meta-analysis. Ecol. Econ. 37, 257–270.
Xie, Z., Wang, Y., Li, F., 2005. Effect of plastic mulching on soil water use and spring wheat yield in arid region of northwest China. Agric. Water. Manage. 75, 71–83.
Yazar, A., Sezen, S.M., Gencel, B., 2002a. Drip irrigation of corn in the southeast anatolia project (GAP) area in Turkey. Irrig. Drain. 51, 293–300.
Yazar, A., Sezen, S.M., Sesveren, S., 2002b. LEPA and trickle irrigation of cotton in the Southeast Anatolia Project (GAP) Area in Turkey. Agric. Water. Manage. 54, 189–203.
Zhang, H., Oweis, T., 1999. Water–yield relations and optimal irrigation scheduling of wheat in the Mediterranean region. Agric. Water. Manage. 38, 195–211.
Zhang, Y., Eloise, K., Yu, Q., Liu, C., Shen, Y., Sun, H., 2004. Effect of soil water deficit on evapotranspiration, crop yield, and water use efficiency in the North China Plain. Agric. Water. Manage. 64, 107–122.
Zhang, B., Li, F., Huang, G., Cheng, Z., Zhang, Y., 2006. Yield performance of spring wheat improved by regulated deficit irrigation in an arid area. Agric. Water. Manage. 79, 28–42.
Zhao, H., Xiong, Y., Li, F., Wang, R., Qiang, S., Yao, T., Mo, F., 2012. Plastic film mulch for half growing-season maximized WUE and yield of potato via moisture-temperature improvement in a semi-arid agroecosystem. Agric. Water. Manage. 104, 68–78.
Zhao, X., Liu, S.L., Pu, C., Zhang, X.Q., Xue, J.F., Zhang, R., Wang, Y.Q., Lal, R., Zhang, H.L., Chen, F., 2016. Methane and nitrous oxide emissions under no-till farming in China: a meta-analysis. Glob. Change Biol. 22, 1372–1384.